O1 — Climate science is accounting

EC2113 Introduction to Climate Science · Module 3 (Oceans) · Lecture 1 · Wednesday 28 October 2026

Some numbers below were on the slides but not said aloud in class (the accounting sum, the 5.7 PW transport, the atmosphere's share of sunlight). They are part of the lecture: learn them from here.

The question

Argha's module was about energy balance. Suhas's module was about air moving around. Are these two separate topics, or one? This lecture argues that they are one: a planet's climate is a set of account books, and the circulation of the atmosphere and ocean is how the books are kept balanced. The ocean is one of the main bookkeepers.

1. Why ask about climate at all? Other planets

In the last decade or so astronomers have found thousands of planets around other stars. "Are we alone?" is a philosophical question, but it can be turned into a scientific one: **which of these planets could keep liquid water at its surface?** Liquid water is the most promising condition for life.

Two things decide how much starlight a planet gets: how hot its star is, and how far away the planet is. A cool star with a close planet can give the same starlight as a hot star with a distant one. The **habitable zone** is the band around a star where a rocky planet can keep surface water liquid. Earth is in it; so, on current estimates, are Proxima b and some of the TRAPPIST-1 planets, and probably Mars. Venus is not.

2. A planet as one dot: the books must balance

The simplest climate model treats a planet as a single dot: sunlight in, infrared out.

At this scale a planet **cannot stay out of balance for long**. If it gained even a few W/m² more than it lost, year after year for a billion years, it would keep heating until its rock melted and boiled away. If it lost more than it gained, it would cool until it reached the temperature of the cosmic background, about 3 K. We see thousands of planets, and they are neither all evaporating nor all frozen dark, so on average planets are close to balance.

There are exceptions, and they help:

- **Jupiter** gives off about twice the energy it absorbs (Argha: ~7 in, ~14 out, W/m²). It is still losing the heat left over from its formation.
- **Earth** also has heat left over from formation, plus heat from radioactive decay, but it has to leak out through a solid lithosphere, which conducts heat very poorly. The geothermal flux is only about **0.09 W/m²**, against about **240 W/m²** of absorbed sunlight: thousands of times smaller. That is why Earth's energy balance can be treated as purely radiative.
- **55 Cancri e** is so close to its star that no solid crust can form on its day side. It is probably a lava world.

3. Why one dot is not enough

From another star, Earth would be a single pixel that brightens and dims as it rotates. It is darkest when the Pacific faces the telescope, because there is so little land in view.

But a single dot hides things that matter. Many planets in the habitable zones of small, cool stars are probably **tidally locked**: like the Moon facing Earth, one side always faces the star. Their *global average* temperature may look fine, while the permanent day side is scorching and the night side frozen, and neither half is habitable. **The global mean can say yes while the map says no.**

Not covered in class. Moving heat from the day side to the night side then decides whether such a planet is habitable at all. Kite and colleagues (2011) showed ways a tidally locked planet inside the habitable zone could still become uninhabitable, depending on how its atmosphere carries heat. Remember Argha's snowball Earth, where one planet can have two stable climates, ice-free and frozen, and jump between them. On tidally locked planets, models suggest that jump doesn't happen, even when a moving ocean is included (Checlair et al. 2019). Transport changes not just *how* warm a planet is but *which kinds* of climate it can have.

4. Earth's map: no single place is in balance

The CERES satellite instruments have measured Earth's top-of-atmosphere radiation for more than twenty years. The map of **net radiation** (sunlight absorbed minus infrared emitted), averaged over ten years, shows:

- a gain of about **+72 W/m²** (zonal mean) in the tropics, roughly between 30°S and 30°N;
- a loss of about **−126 W/m²** near the poles.

A thought experiment in accounting. Suppose all that tropical gain went into the top 10 m of an ocean and nothing carried it away. After 20 years, the equator would have warmed by about **1,100 K**, hot enough to melt rock. The polar loss would take 10 m of water down past absolute zero in about **3 years**. Neither has happened, and temperatures have hardly changed over those 20 years. So the energy must have been **moved**: from where it is gained to where it is lost.

(This is a question about where the energy went, not about what would happen if transport stopped. If transport stopped, temperatures would change and so would the radiation, and each latitude would settle at its own balance. You'll try both in the tutorial.)

How much must move? Adding up the imbalance from the pole gives the energy that has to cross each latitude. It peaks at about **5.7 PW** (petawatts, 10^{15} W) near 35° in each hemisphere. That is the job of the atmosphere and the ocean together.

5. Moving energy means moving mass and momentum

Mass. Air (or water) carries the energy. If air kept flowing out of the tropics towards the poles, the tropics would run out of air. So mass must balance too: what goes poleward must come back. That closes a loop, which is the basic idea of the Hadley cell.

Angular momentum. On a rotating sphere, each parcel of air also keeps its angular momentum. Near the equator, air is a long way from Earth's axis and moving fast with the surface. As it moves poleward it gets closer to the axis, and so it moves faster and faster *relative to the ground* beneath it, even though no force pushes it east. That apparent eastward push is what we call the Coriolis effect: it exists only because we watch from the rotating ground. If Earth were a cylinder, the distance to the axis would not change with latitude, and there would be no such effect.

If air kept its angular momentum all the way, winds would reach about **130 m/s** in the subtropics. We don't see that, because such strong wind shear is unstable: the flow breaks up, and the Hadley cell ends.

- On a **slowly rotating** world such as Saturn's moon Titan, one Hadley cell can reach almost to the poles.
- On a **fast-rotating** planet, energy has to be passed poleward in shorter steps, so we expect more cells.

Momentum is exchanged with the solid Earth. Where surface winds are westerly, air moves faster than the ground and friction passes momentum to the Earth. Where they are easterly (the trade winds), the air gains momentum from the Earth. The atmosphere does not balance its momentum budget alone. The exchange is real enough to measure: the length of Earth's day varies slightly over days to weeks as air and water shift around.

Why a fluid at all? The Moon has no atmosphere or ocean. The only way to move heat across it is conduction through rock, which is very slow, so its day side is very hot and its night side very cold. It takes a moving fluid to even out a planet's temperatures.

So energy, mass and momentum each have a ledger, and climate science is keeping the books. *The best climate scientists are accountants.*

6. "It depends": every answer has a scale

If the atmosphere moves energy, mass and momentum so well, why study the ocean, or ice sheets, at all?

Because the climate system changes on many timescales at once. The atmosphere changes over days to weeks. The ocean, land, vegetation and ice change over years, decades, centuries and millennia: El Niño (a strong one is under way in the Pacific this year), the ice ages, Milankovitch cycles. The slower parts set the conditions the fast parts respond to. In the Eocene, about 50 million years ago, there was almost no ice anywhere, and crocodile-like reptiles lived in the Arctic. An ice sheet looks still, but it changes the albedo, the heat capacity and the fresh water that the atmosphere and ocean respond to.

So whenever someone asks "is X important for climate?", the right first words are **"it depends"**: on the space and time scale. Newton's laws work the same for an atom and a galaxy. The "laws" of climate science do not; they depend on the scales you study. For the ice ages, you can almost ignore the atmosphere and focus on ice and albedo. For next week's weather, you can ignore the ice sheets.

7. The ocean's ledgers

In this module we keep the ocean's books:

Ledger	Week
Energy	1
Momentum	2
Mass and salt	3
Carbon (and N, P, Fe, Ca, which life in the ocean needs)	4

For each one we wear three hats:

1. **Accountant:** which budget, and what goes in and out?
2. **Data scientist:** what does it look like in the data?
3. **Physicist (or biologist):** why does it look like that?

8. First entry: who absorbs the sunlight?

Of the ~**242 W/m²** of sunlight Earth absorbs (global mean), about **78** is absorbed in the atmosphere (about a third) and about **164** at the surface. The ocean covers 72% of the surface but takes about 77% of what the surface absorbs: **roughly half of all the sunlight Earth absorbs goes into the ocean.**

The absorption map has clear patterns. Asking *why* of a pattern like this is where the interesting questions start, and some of them still don't have precise answers.

Next lecture: where does the ocean's energy go?

Readings

Core

- Hartmann, *Global Physical Climatology* (2nd ed., 2016), §2.8 "The Energy Balance at the Top of the Atmosphere" and §2.9 "Poleward Energy Flux", pp. 40–48 (PDF pp. 56–64). The CERES maps and the transport curve from this lecture.

Optional

- Hartmann §6.4 "The Angular-Momentum Balance", pp. 176–180 (PDF pp. 192–196): a recap of Suhas's angular-momentum argument.
- Hartmann §1.7 "The World Ocean", pp. 17–19 (PDF pp. 33–35): the basic numbers of the ocean.
- For the curious (research papers, not core): Kite, Gaidos & Manga (2011), "Climate instability on tidally locked exoplanets", *Astrophysical Journal* 743, 41; Checlair et al. (2019), "No snowball bistability on synchronously rotating planets even with a dynamic ocean", *Astrophysical Journal Letters* 884, L46.
- Stewart, *Introduction to Physical Oceanography* (2008), §5.6 "Geographic Distribution of Terms" and §5.7 "Meridional Heat Transport", pp. 65–70 (PDF pp. 73–78). A preview of next lecture.

Questions to think about

1. Why does an imbalance make sense for a place on Earth but not for Earth as a whole?
2. A planet's global-mean temperature is 15 °C. Can you say it is habitable? What else would you want to know?
3. Would a planet that spins faster find it easier or harder to move heat to its poles? Why?
4. Name one question about climate where the ocean doesn't matter, and one where it is all that matters.